Protein-guided nucleotide viral genome assembly for huge metagenomic datasets

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Abstract

Viruses are the most abundant entities on earth. Metagenomic studies offer an unique opportunity to detect them. Their genomes can be assembled by tying together short nucleotide sequence reads into continuous genomic stretches. However, their high mutation rates and large viral communities make this particular challenging for assemblers.

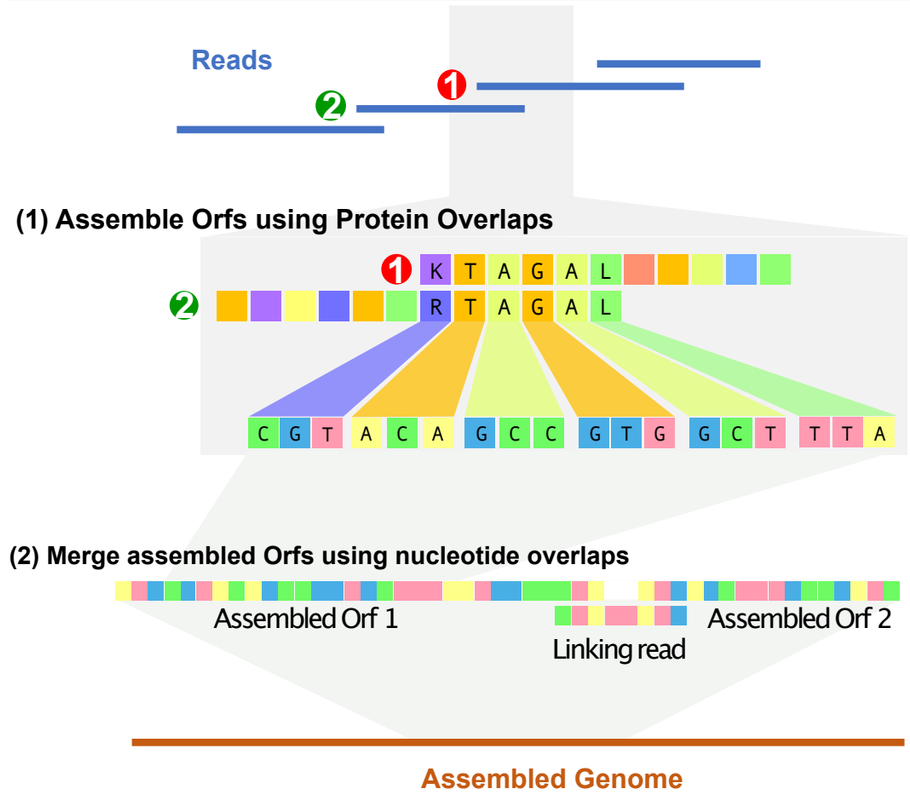
With our new protein-guided nucleotide assembler (**PenguiN**) we are able to assemble long contigs and whole viral genomes. On a simulated metagenome, we obtain a **3 to 11 fold increase** in sensitivity compared to MEGAHIT at similar precision. On a metatranscriptomic dataset from 82 sludge samples, **PenguiN** assembles about **75-90% more complete** ssRNA phage genomes than the state-of-the-art tools.

Code availability

PenguiN is GPLv3-licensed open source software

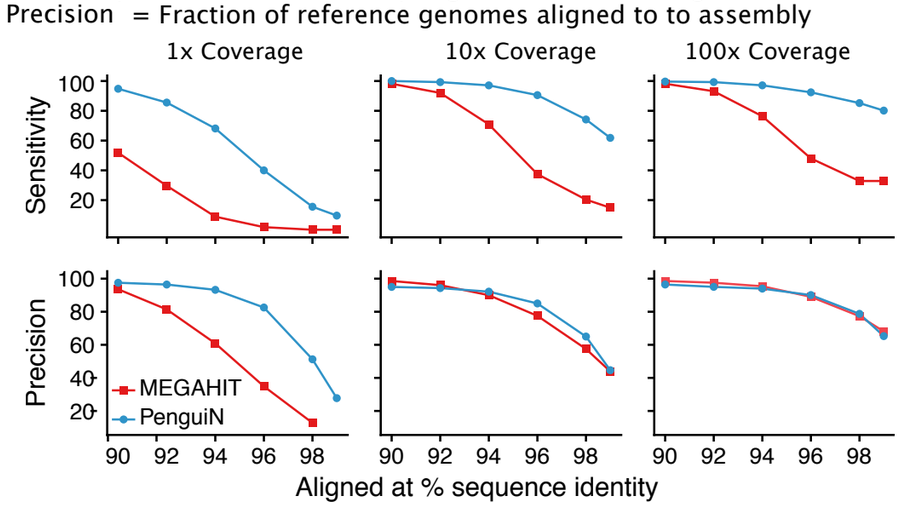
plass.mmseqs.com

Protein-guided assembly



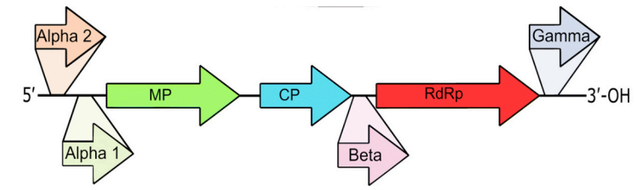
Assembly of 2550 HIV genomes from simulated reads at 1x, 10x, 100x

Sensitivity = Fraction of assembly aligned to a reference genome
Precision = Fraction of reference genomes aligned to to assembly

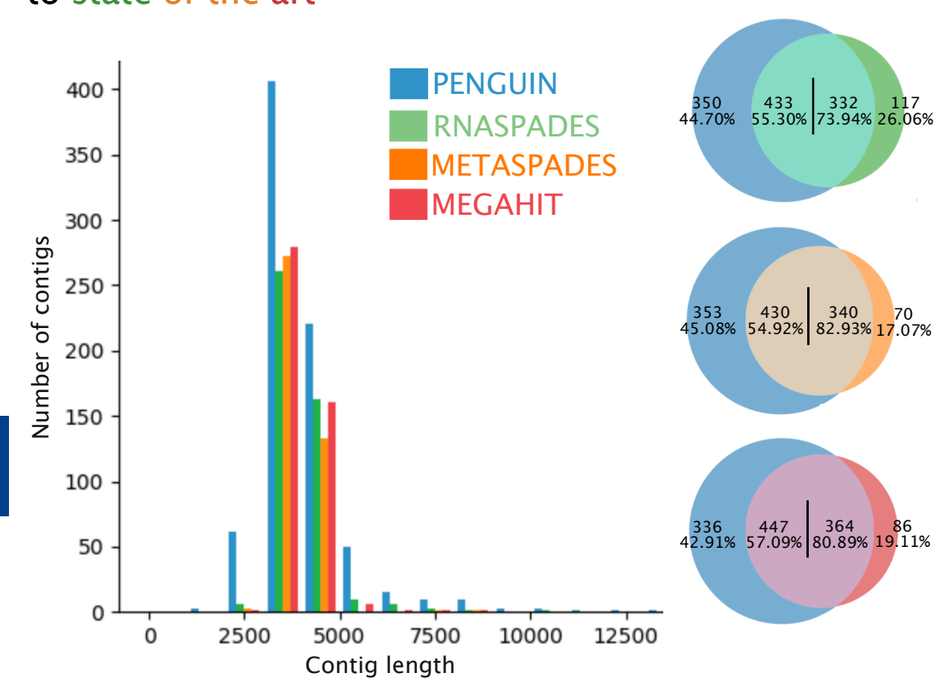
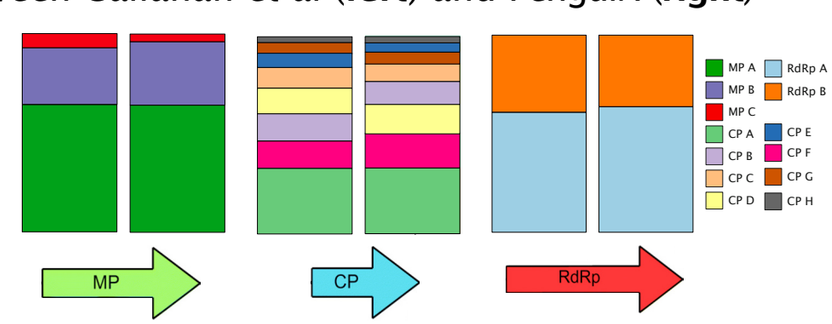


Assembly of ssRNA phage genomes

ssRNA phage genomes consists of three core genes in different variations (Callanan et al)



Metatranscriptome assembly of ssRNA phages. **PenguiN** assembles **many fold** more phages compared to state of the art



Protein distribution comparison of 3 core proteins between Callanan et al (**left**) and PenguiN (**right**)